Design, Control, and Experimental Validation of Highly Dexterous Multimodal Rover Leg

Background and Significance

Planetary exploration relies on rover mobility, as the ability to traverse varied terrain determines the range and quality of achievable science [1]. Extraterrestrial surfaces present significant variability, from fine granular soils such as sand and gravel to coarse rocks and boulders [2, 3]. Fixed mobility systems perform inconsistently across this spectrum, resulting in excessive slip, elevated energy consumption, and reduced travel speeds. In severe cases, immobility can terminate missions, as demonstrated by NASA's Spirit rover, which became trapped in Martian soil in 2009 before completing its objectives [4, 5]. Adaptive mobility offers a solution by allowing changes in wheel geometry, limb posture, or suspension in response to terrain. Such systems improve efficiency in typical environments while reducing entrapment risk in challenging ones. Contemporary research in adaptive mobility, for instance, has produced multimodal and bio-inspired platforms capable of reconfiguring their morphology to sustain performance under unpredictable conditions [6, 7, 8, 9].

Despite these advances, no wheeled platform has demonstrated the capability to transition between multiple dexterous bio-inspired architectures, and the pursuit of an ideal rover platform remains an open question in academia. To address this gap, as a part of the AstroX I am developing
a multimodal rover capable of transforming between mammalian and arachnid configurations, as outlined in Figure 1. The proposed design enables architectural transformations while preserving high dexterity in each mode, supporting faster and more reliable locomotion across both unconsolidated and consolidated terrains.

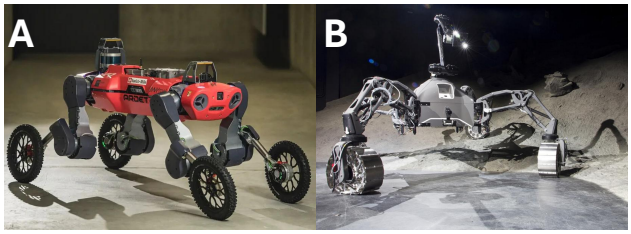


Figure 1: Examples of adaptive wheeled-legged rover platforms. (A) ANYmal with integrated mammalian wheeled legs, demonstrating transitions between walking and driving modes [10]. (B) SherpaTT in an arachnid-inspired configuration with articulated limbs for rough terrain navigation [11].

Specific Aims, Goals, or Questions

This project examines how extreme multimodality can enhance planetary exploration. Within the semester timeframe, the scope is limited to a single leg rather than a full rover platform. Demonstrating the benefits of multimodality at the leg level will motivate future system-scale work, and with considerable prior systems engineering already completed, a full 7-DOF leg is achievable within the semester. The specific objectives are as follows:

- Design and manufacture torque-transparent low gear ratio actuators for robust, precise, and natural limb articulation.
- Integrate these actuators into a wheeled rover leg using parallel mechanisms to achieve a highly responsive, low-inertia design capable of dexterous motion.
- Evaluate the developed leg on a 3-axis 8020 gantry on the AstroX's
planetary testbed to assess performance metrics in an analog planetary environment.

Approaches

Actuator Design: We will develop a custom 20 N·m, low gear-ratio actuator to enable high mobility with torque transparency and shock resilience. The design combines a permanent magnet synchronous motor (PMSM) with a custom 18:1 two-stage planetary gearbox engineered for low backlash and durability. Power transmission and coupling elements will use robust metal components, with shafts and planet carriers machined in Georgia Tech's MMM Hall and enclosed in an FEA-optimized housing. Position feedback will be provided by hall-effect magnetic encoders, while current-controlled field-oriented control (FOC) drivers such as the ODrive S1 will deliver precise torque and joint control.

7-DOF Low Inertia Leg Design: In parallel with actuator development, a 7-DOF wheeled leg will be designed and manufactured to enable transformation between target configurations and maintain multimodality. As shown in Figure 2.A, the general architecture supports these transitions, while the use of parallel differential mechanisms (Figure 2.B) reduces limb inertia for greater responsiveness and energy efficiency. Structural elements will be CNC-machined or assembled with robust mechanical connectors to ensure low deflection, precise articulation, and durability. The leg and actuators will integrate to form the complete articulated system for further testing.

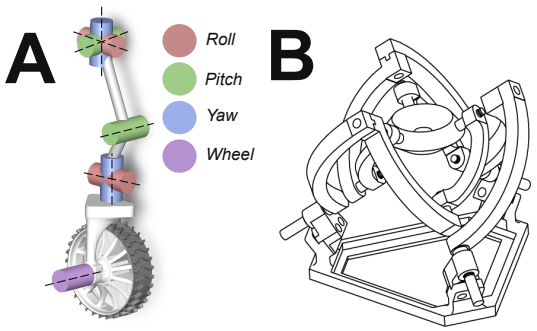


Figure 2: (A) 7-DOF wheeled leg architecture diagram with roll, pitch, and yaw joints color-coded. (B) 3-DOF parallel mechanism for reduced lower-leg inertia.

Leg Performance Evaluation: The completed leg will be mounted on a 3-axis 8020 test stand that allows linear motion in x, y, and z while constraining rotation, replicating rover chassis conditions. High-precision linear encoders, current sensors, and hall-effect encoders will measure wheel slip, joint torques, and end-effector forces derived from the Jacobian. Additional metrics, including speed and cost of transport, will be evaluated using these sensors. Testing will take place on the AstroX's analog lunar testbed to provide a realistic assessment of performance across gradients and granularities.

Outcomes

By the conclusion of this project, sufficient data will be collected on the proposed leg architecture to determine whether extreme multimodality enhances extraterrestrial navigation. This outcome will lay the groundwork for developing a four-legged multimodal rover system while also contributing valuable insights to the broader field of space robotics.

Beyond the technical scope, this project marks an important step in my growth as both an engineer and a researcher. There is still much to learn, build, and explore in robotics, and that challenge excites me. This work gives me the opportunity to design a robust, machined platform that requires advanced mechatronic skills, pushing me to grow by turning scientific ideas into practice. As I finish my B.S. in Fall 2026, this project helps me prepare for a Ph.D. in space robotics by giving me a stronger technical and scientific foundation to advance robotic mobility and contribute insights that may bring the stars closer to humanity.

References

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Student Responsibilities and Commitments

Weekly Time Commitment: I will only register for 12 credit hours of light coursework in the Spring 2026 semester, with no lab block classes. This, along with my minimal extracurricular commitments, will let me dedicate a significant amount of time to this project. With all this time, I plan to dedicate 20 hours a week towards research. I hope to commit to going into lab three days a week and working 6-7 hours while there.

Methods and Frequency of Communication: My direct mentor, Dr. Nakka, and I agreed that communication will involve weekly in-person meetings and frequent online discussions over the lab's Slack server. Dr. Nakka is committed to supporting my development as a student researcher and has allocated 1-2 hours weekly for individual discussions regarding project updates, my inquiries, and his own feedback. Communication via the lab's Slack chat is mainly to discuss current literature, design challenges, and immediate questions that he or I may have.

Data Recording, Storage, and Sharing: The lab's GitHub Enterprise will store and organize all research data, including CAD models, Python scripts, and CSV files, enabling version control and easy access. I will make weekly updates to the repository before each lab meeting, ensuring an up-to-date representation of the ongoing work. All pertinent information will be reflected in a continuously updated PowerPoint meant for the weekly meetings.

Reflection on Successes and Challenges: During weekly meetings, I will reflect on my work with Dr. Nakka. We agreed that it is important to outline objectives with specific deadlines, and I will compare project progress against these goals while identifying areas that may need adjustment and addressing challenges. Dr. Nakka emphasized that discussions will also focus on learning from mistakes and examining why they occurred. Having already worked together for over a year, we know that we collaborate well, which gives me confidence that these meetings will be productive and constructive.

Mentoring Team Responsibilities and Commitments

Roles of the Mentoring Team: Dr. Nakka, our lab's principal investigator, intends for this project to be a learning experience that helps me grow as both an engineer and a researcher. His role is to provide high-level guidance and feedback, plan scientific objectives, and assist with writing and reviewing the paper we aim to publish from this work.

Training Plans: My freedom within the project means that I will be leading the rover design, manufacturing, and experimental setup. The goal is to give me room to learn, as I will be involved in the entire experimental process. Although I have a lot of autonomy, Dr. Nakka has expressed that he will always be available to talk to for feedback and support if I have any questions or concerns. Dr. Nakka has also stated that he will consistently make an effort to provide me with resources that he deems relevant to my research, such as recent publications or textbooks.

Funding Sources for Materials and Supplies: The funding for stock, machining, and all other expenses will come from the lab's budget. Since the AstroX
is relatively new, Dr. Nakka and I have talked about how the project's spending on 3D printers and terrain testbeds will aid in the development of future projects, as Dr. Nakka has a continued interest in prototyping platforms for interaction with varied media.